

Linear systems and elimination

# Sharp row-deletion singular-value limit for spherical random matrices

**Difficulty:** challenging **Importance:** interesting to the community

**Status:** open in the checked literature

## Problem statement

Fix  $0 < \theta < 1$ . For  $A \in \mathbb{R}^{m \times n}$  define

$$s_\theta(A) = \min_{S \subseteq \{1, \dots, m\}, |S| = \lfloor \theta m \rfloor} \min_{\|x\|_2=1} \|A_S x\|_2,$$

where  $A_S$  contains the rows indexed by  $S$ . Let  $a_\theta > 0$  satisfy  $\mathbb{P}(|G| \leq a_\theta) = \theta$  for  $G \sim N(0, 1)$ , and set

$$h_\theta = \frac{1}{\sqrt{2\pi}} \int_{-a_\theta}^{a_\theta} t^2 e^{-t^2/2} dt.$$

For every integer sequence  $(m_j, n_j)$  with  $n_j \rightarrow \infty$  and  $m_j/n_j \rightarrow \infty$ , let the rows of  $A_j \in \mathbb{R}^{m_j \times n_j}$  be independent and uniformly distributed on  $\mathbb{S}^{n_j-1}$ . Is

$$\frac{s_\theta(A_j)^2}{\|A_j\|_2^2} \xrightarrow{\mathbb{P}} h_\theta?$$

This states Steinerberger’s proposed asymptotic with  $\theta = q - \beta$ . Convergence in probability and the floor convention make the source’s limiting statement explicit. The source also asks for quantitative error bounds; these belong to this problem rather than separate entries.

## Connection to numerical linear algebra

The quantity measures the worst conditioning remaining after rows are discarded. It controls convergence guarantees for quantile Kaczmarz methods on corrupted linear systems.

## References

1. S. Steinerberger, *Quantile-based Random Kaczmarz for corrupted linear systems of equations*, Information and Inference **12**(1) (2023), 448–465, §2.3, equation (8) and the following paragraph. [Published paper](#). [Author preprint](#), v1 (2021), §2.3, equation ( $\diamond$ ).
2. E. Battaglia, J.-F. Cai, J. Chen, A. Ma, D. Needell, and T. Wu, *Quantile Randomized Kaczmarz for Streaming Linear Systems with Massart Noise*, arXiv:2608.27968v1 (2026), §1.1, discussion of Steinerberger’s random-matrix heuristic and the distinction between static and streaming data; §5. [Preprint](#).

## Status check

Checked the published formulation and the arXiv version history (2107.05554 has only v1), then searched the problem’s title, the author’s name with “subsingular”, and combinations of “trimmed/smallest singular value”, “quantile”, “Kaczmarz”, “heuristic”, “proof”, “2025”, and “2026”. The August 2026 streaming paper still describes the original estimate as a heuristic; its fresh-sample convergence bounds do not establish this static-matrix limit. No later proof or counterexample was located in this bounded search. That is a literature-status assessment, not a proof of openness.